

MURALI SAI BUDDAKKAGARI VENKATA

AI/ML ENGINEER

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EXPERIENCE

Querkey Inc

Jun 2025 - May 2026

Data Science Intern

CA, US

- Gathered requirements from healthcare clients in sprint planning and review sessions, then developed and deployed a proof-of-concept real-time Patient Flow Tracking System using **multimodal AI** (Gemini 2.5 Flash) with **chain-of-thought** and **few-shot prompting**, integrating a **RAG** pipeline that structured unstructured patient notes for real-time insight retrieval.
- Cut **production** ML workflow failures 60% across 100K+ records by deploying a **Dockerized** observability stack (MLflow, **FastAPI**, ChromaDB, HuggingFace monitoring), enabling real-time failure detection.
- Authored a 30-query **SQL** bank to ground an AI chatbot's **RAG**-based answers to natural-language questions, and built a Superset **REST API** integration generating dashboard charts on demand, demoing both to non-technical stakeholders with plain-language visuals to confirm they met their needs.

VIEW GROUP INDIA (FORMERLY VIEW SYNERGY)

Jan 2024 - Aug 2024

AI and Automation Intern

KA, IND

- As part of a cross-functional project team modernizing a live production platform, reviewed React frontend, .NET **Azure** Functions **API**, and MySQL stored-procedure code, tracing 2 recurring exceptions to root cause and surfacing SQL-injection vulnerabilities that fed into a 94-issue security assessment.
- Built and tested a new Roles/Designations CRUD feature end to end, including new MySQL stored procedures, a .NET **API** layer, and a React frontend, while documenting Microsoft Entra ID role setup for a proposed identity migration.
- Audited Redis usage across all 6 backend **microservices** and documented Microsoft Entra ID app registration and role setup, supporting a proposal to replace a fully custom JWT/BCrypt authentication system, and presenting progress to finance-domain clients in sprint reviews with plain-language visuals.

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Deployed a **production** financial **document intelligence** RAG system, not a demo, by building an end-to-end **data pipeline** (EDGAR download, section parsing, chunking, **embeddings**) into 8,232 SEC 10-K chunks across 5 companies, served by a **Python/FastAPI** service with ChromaDB, cross-encoder reranking, **RBAC**-enforced information barriers (FINRA Rule 2241 / SEC Rule 15g-1), and a **Dockerized** serverless Google Cloud Run deployment verified by a 504-test **CI** suite.
- Built a 54-question **LLM evaluation framework** (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness) to measure a bounded **agentic retry** (**tool calling** with an iteration cap, failure abort, and fail-closed fallback) and **prompt-injection** / MNPI **guardrails**, and documented 17 diagnosed failures and fixes in a written case study and Streamlit dashboard, solo-built using **AI-assisted engineering** practices (**Claude Code**).

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Shipped a **production agentic workflow** system (19 **REST** endpoints) that plans and executes multi-step tasks end-to-end, validated by 106/113 passing tests against a live deployment, by architecting a 3-layer **LangGraph** hierarchy (planning supervisor, parallel specialists, reflective reviewer) with a rework loop, multi-model routing (GPT-4o, Claude Haiku), and two-tier **Redis** + Supabase **pgvector** memory.
- Built a custom **MCP** (Model Context Protocol) tool server integrating internal and third-party **APIs** across 6 **agent skills** with per-agent tool permissions and rate limits, instrumented end to end with **OpenTelemetry** distributed tracing and per-task cost tracking, and a **Human-in-the-Loop** system escalating across 4 severity levels with full audit-trail logging, independently architected using **spec-driven development** practices in **Claude Code**.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (React), SQL

AI & Agent Engineering: LangChain, LangGraph, LangSmith, LLM APIs (OpenAI, Claude, Gemini), **Agentic Workflows & Tool Calling (MCP)**, Multi-Agent Systems, **RAG**, Prompt Engineering, **LLM Evaluation (RAGAS)**, Document Intelligence, Guardrails & Failure Handling, Statistical Evaluation & EDA, Observability (**OpenTelemetry**), Embeddings, Vector Databases, HuggingFace, PyTorch, TensorFlow

Cloud & Infrastructure: Docker, FastAPI, REST APIs, Microservices, Google Cloud Platform, Azure, CI/CD, GitHub Actions, Test Automation, MLflow, Redis, PostgreSQL, ChromaDB, Git

EDUCATION

Stevens Institute of Technology

May 2026

Master of Science, Applied Artificial Intelligence

NJ, US

PES University

May 2024

Bachelor of Technology, Computer Science (Specialization in ML)

KA, IND

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens Institute of Technology (Advised by Prof. Kevin Lu). A size-conditioned Weighted Box Fusion ensemble of **deep learning** detectors (Faster R-CNN, Mask R-CNN, and YOLO26) for cell-nucleus detection, reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl benchmark.
- Presented at ICICC 2024 and won Best Research Paper Award - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.